

ILLUSTRATION BY **RAJ VERMA**

power, which is about 2% of electricity generation. The country needs to pull forward its 100 GW nuclear energy target—30 GW by 2035, 60 GW by 2040—with pre-approved sites, standardised reactor designs, and a dedicated financing vehicle.

India's decisive advantage are thorium reserves of 650,000 tonnes—25% of global deposits. The global Small Modular Reactors (SMR) market is projected at \$300 billion by 2040. India has the engineering talent, the manufacturing

scale, and the resource foundation to build and export SMRs across Southeast Asia and Africa—a \$50 billion industry waiting to be claimed.

### RENEWABLES DEMANDS

India's renewable resource endowment is extraordinary—3 TW of potential, more than China and Europe combined. Rajasthan alone holds 142 GW of exploitable solar. Offshore wind along 7,500 km of coastline adds 127 GW more. The sun is not the problem in generating

solar; the supply chain, the grid, and the storage behind it are. Until those are fixed, India is not building energy sovereignty.

India still imports the majority of solar cells from China and has zero domestic polysilicon refining capacity. Building the full stack—from polysilicon to wafers to cells to modules to batteries—requires \$15–20 billion over five years and creates an export base for a Global South actively seeking alternatives to Chinese supply.

### MINERALS RACE

Energy transition runs on minerals—lithium for batteries, cobalt for cells, nickel for anodes, rare earths for turbines and motors. China controls 60–90% of processing capacity for every one of them.

India holds 6.9 MT of rare earth reserves—the third largest globally—yet contributes less than 1% of global production. Its 2023 discovery of 5.9 MT of lithium in Jammu & Kashmir is among the world's largest. We must fast-track the J&K lithium project and build mineral processing zones.

### STRATEGIC IMPERATIVE

What India must now do is treat energy exactly as it treated defence. From importing nearly 70% of its requirements a decade ago, India is today targeting \$25 billion in defence exports. The transformation did not happen by accident; it did because India decided it would. Energy sovereignty demands the same conviction. **BT**

*Read full version on: <https://www.businessstoday.in/opinion/columns/story/viksit-bharat-will-need-energy-sovereignty-not-just-energy-security-523702-2026-04-02>*

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